

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (CE/IT)

May 2014

IT405 Advanced Computing

Date: 24.05.2014, Saturday

Time: 01:30 pm To 04:30 pm

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) What is cloud computing? Explain How multi core processor have made cloud computing efficient. [05]
- (b) Write the limitations of cloud computing application. [02]
- Q - 2 (a) What is hardware assisted virtualization? What are the differences between Hardware assisted virtualization, Full virtualization and Para virtualization? [07]
- (b) "Cloud computing is not always secure". Justify the statement. List the application of cloud computing in the field of engineering. [07]

OR

- Q - 2 (a) Explain the Xen hypervisor in detail. [07]
- (b) Explain the para virtualization and full virtualization.. Which of them provides superior performance? List the pros and cons of para virtualization and full virtualization. [07]
- Q - 3 (a) What is cluster? List the steps to install MPICH cluster master node on Ubuntu node. [07]
- (b) What is Google App Engine? List and explain the functionality provided by Google App Engine? [07]

OR

- Q - 3 (a) Describe the working of Hadoop. Explain the various elements of Hadoop distributed computing environment. [07]
- (b) What is Big Data? Write a note on services provided by amazon. [07]

SECTION – II

Q - 4 Define grid computing and explain its components with an architecture diagram in detail. [07]

- Q-5** (a) Differentiate between cluster, grid and cloud computing in detail. [07]
(b) Explain briefly Resource Scheduling techniques in cluster computing. [07]

OR

- (b) Compare and contrast HPVM (High Performance Virtual Machine) and CLUMPS (Clusters of SMP). [07]

Q-6 (a) What is cluster computing? List the factors to be considered while implementing cluster. [07]

- (b) What is virtual organization in grid computing? Explain resource orchestration. [07]

OR

- (b) Explain how Globus tool kit provide framework for implementation of grid. List the various packages available in Globus tool kit [07]
